CARBOHYDRATES

- Sugars and sugar containing drugs
- Compounds related to sugars
- Polysaccharides
- Gums and mucilages

Introduction

- Carbohydrates are produced in green plants in the presence of chlorophyll and sunlight in a process known as photosynthesis.
- They serve as food sources for living organisms and provide the structural support for plants.
- Many carbohydrates are large polymers composed of repeating units of simple sugars.

Introduction

- The word carbohydrate can be traced back to Germans, who called them '*kohlenhydrates*'.
- It was then termed carbohydrates in English.
- Carbohydrates are naturally occurring polyhydroxy aldehydes or polyhydroxy ketones or compounds hydrolyzed to give them.
- The formulas of many carbohydrates can be written as carbon hydrates, $C_n(H_2O)_n$

Origin of Sugars

- Sugar is generally extracted from plants like sugar beet and sugarcane or found in a variety of different foods.
- Honey, for example, contains a high amount of sugar which is why it is sweet.
- Sugar is a main ingredient in many foods and recipes, available as sucrose or saccharose (household sugar), lactose (milk sugar), maltose (malt sugar), fructose (fruit sugar) and glucose (dextrose).

Definition

- In most cases, carbohydrates are composed of the elements C, H, and O where the ratio of H to O is 2:1 as in water.
- They are the most abundant organic compounds in plants rather than in animals.
- Carbohydrates are primary metabolites.
- They are products of photosynthesis.

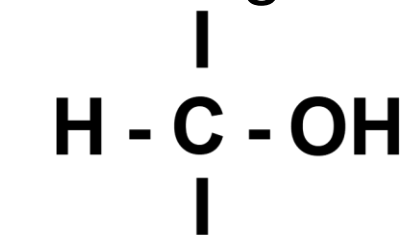
$$n\,CO_2 + n\,H_2O + \text{energy} \longrightarrow C_nH_{2n}O_n + n\,O_2$$

Role of Carbohydrates

- They give rise to all other natural products
- Some of them could be used as taxonomic markers
- They occur combined with non- Carbohydrate plant constituents
- They are extensively used as
  - Food
  - In pharmaceuticals as additives
- Many of them have pharmacological activity
  - Anticoagulants e.g. Heparin
  - Hypoglycemic activity e.g. Guar gum
  - Anti-inflammatory activity
  - Immunomodulator activity
  - Antimicrobial/antiviral
- They can serve as the starting material in the synthesis of important drugs

Types of Carbohydrates

Carbohydrates have the following basic composition:

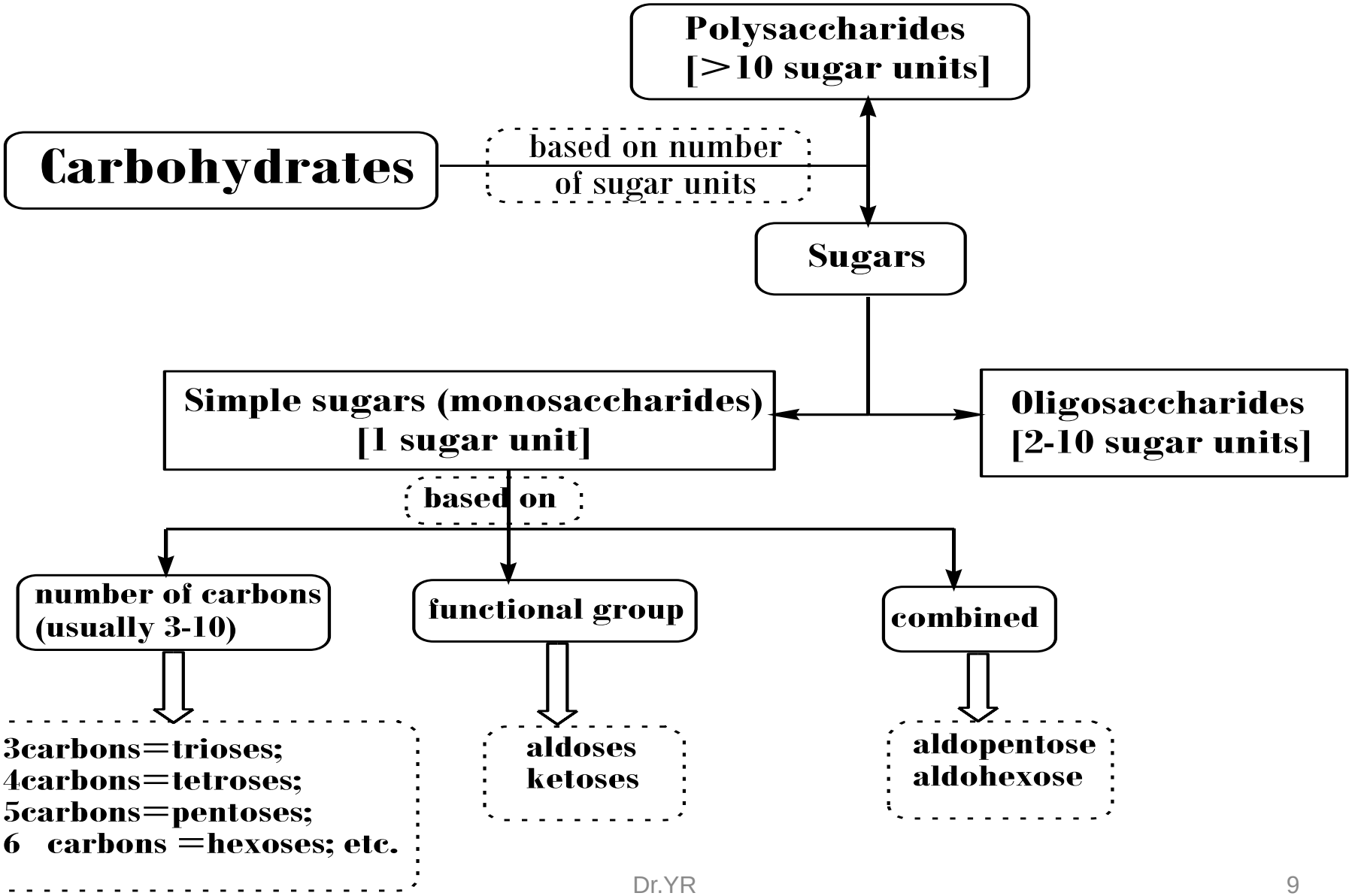
$(CH_2O)_n$ or 

- **Monosaccharides - simple sugars with multiple -OH groups. Based on number of carbons (3, 4, 5, 6), a monosaccharide is a triose, tetrose, pentose or hexose.**
- **Disaccharides - Two monosaccharides linked by a covalent bond.**
- **Oligosaccharides - a few monosaccharides linked by covalent bonds**
- **Polysaccharides - polymers consisting of chains of multiple monosaccharide or disaccharide units.**

Classification of carbohydrates

- Carbohydrates can be divided as
  - –Sugars
    - Simple sugars (monosaccharides)
    - oligosaccharides
  - –Polysaccharides

Classification



Classification of carbohydrates

- **Sugars**

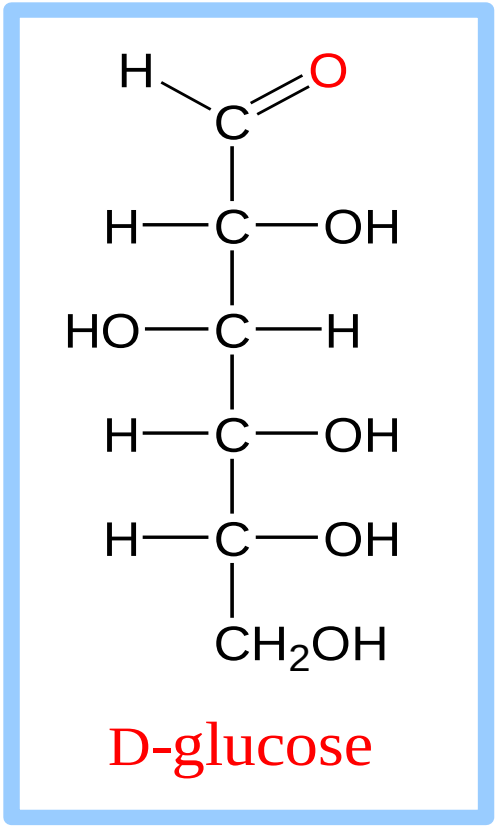
Sugars can be subdivided in to:

- **A. Monosaccharides**, which

1. Are the simplest sugars that cannot be hydrolyzed into smaller units
2. Are monomeric units (building blocks) for complex carbohydrates
3. Could be composed of 3-9 carbons
4. Contain C, H, and O in a 1:2:1 ratio
5. Quick energy sources

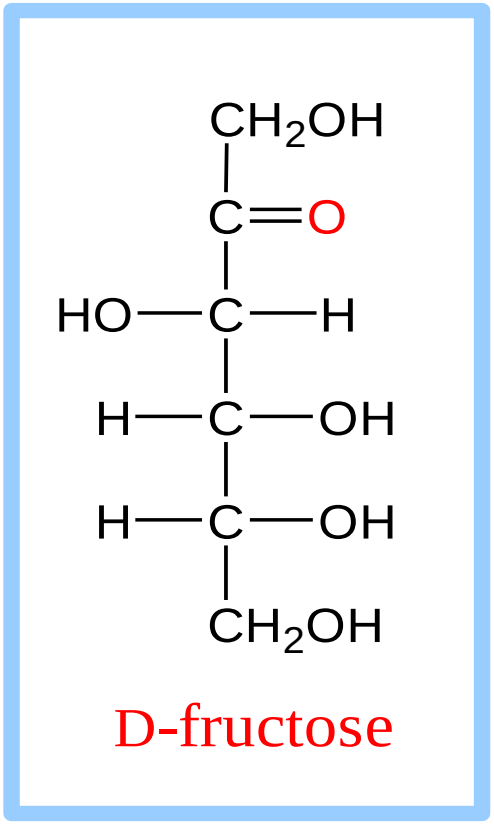
Monosaccharides

Aldoses (e.g., glucose) have an aldehyde group at one end.



D-glucose

Ketoses (e.g., fructose) have a ketone group, usually at C2.



D-fructose

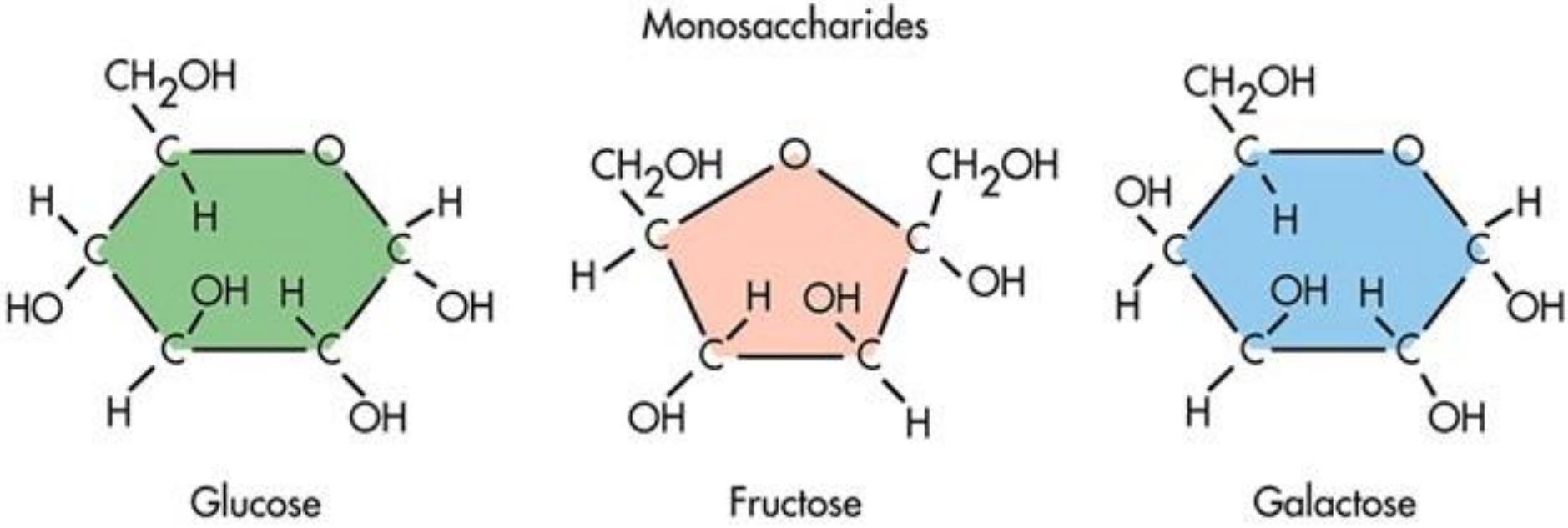
Monosaccharides

Examples:

Glucose $C_6H_{12}O_6$

Fructose $C_6H_{12}O_6$

Galactose $C_6H_{12}O_6$



Classification of carbohydrates

–B. Disaccharides,

- Which upon hydrolysis yield two molecules of monosaccharides are called as disaccharides.

Sucrose $\xrightarrow{\text{Hydrolysis}}$ glucose + fructose(sugarcane)

Maltose $\xrightarrow{\text{Hydrolysis}}$ glucose + glucose(malt sugar)

Lactose $\xrightarrow{\text{Hydrolysis}}$ glucose +galactose(cow's milk)

Sucrose	-	Sugar
Maltose	-	Fermentation & Alcohol production
Lactose	-	Milk products

Disaccharides

- Double sugars
- Good source of energy
- Break down into simple sugars

Sucrose (glucose + fructose)
Lactose (glucose + galactose)

Other **disaccharides** include:

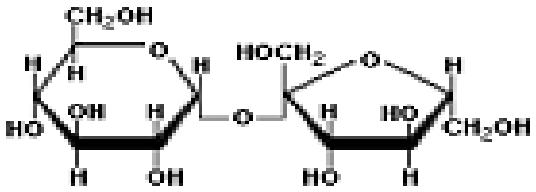
-- Sucrose, common table sugar, has a glycosidic bond linking the anomeric hydroxyls of glucose & fructose.

-- Because the configuration at the **anomeric** C of glucose is **α (**O points down from ring), the linkage is α(1→2). The full name of sucrose is α-D-glucopyranosyl-(1→2)-β–D-fructopyranose.)

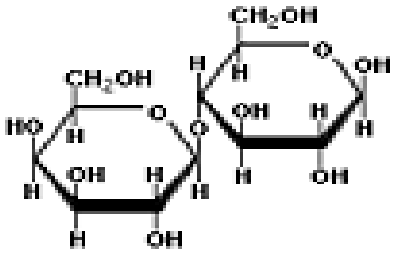
-- Lactose, milk sugar, is composed of galactose & glucose, with β(1→4) linkage from the anomeric OH of galactose. Its full name is β-D-galactopyranosyl-(1→ 4)-α-D-glucopyranose

Disaccharides

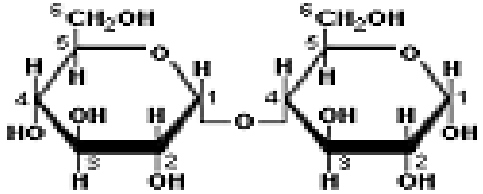
- Compare the structures of these three common disaccharides



Sucrose



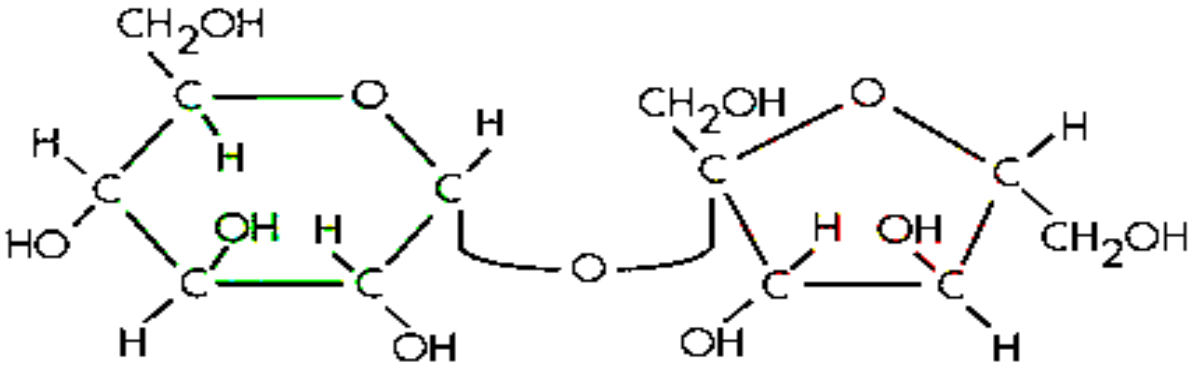
Lactose



Maltose

Disaccharides

Sucrose: glucose + fructose
Lactose: glucose + galactose
Maltose: glucose + glucose



Sucrose

- Sucrose is an α (1-4) link between D-Glucose and D-Fructose
- Lactose is an α (1-4) link between two D glucose

Classification of carbohydrates

—C. Trisaccharides,

- —As the name indicates, these liberate three molecules of monosaccharides on hydrolysis

Raffinose —Hydrolysis→ glucose + fructose+ galactose(in beet and manna)

Gentianose —Hydrolysis→ glucose + glucose+ fructose (gentian root)

—D. Tetrasaccharides

- — Stachyose, a tetrasaccharide, yields on hydrolysis, four molecules of monosaccharides

—E. Polysaccharides,

- —On hydrolysis they give indefinite no. of monosaccharides

Carbohydrates – Properties

- **SUGARS**
  - ✓ are colorless crystalline compounds
  - ✓ have sweet taste
  - ✓ are soluble in water
  - ✓ have low molecular weight
    - e.g. glucose, fructose, and sucrose.
- **POLYSACCHARIDES**
  - ✓ are amorphous solids
  - ✓ are tasteless
  - ✓ are generally insoluble in water
  - ✓ have high molecular weight
    - e.g. starch, cellulose, gums, pectins, inulin etc.

Sugars and sugar containing drugs

- Sugar is the collective name used for naturally occurring carbohydrates. The most common forms are:
  - sucrose – the most commonly occurring and also known as table sugar due to its use in hot drinks. It is derived mainly from sugar cane and sugar beet.
  - lactose – which occurs in milk
  - fructose – which is found in fruit

Sugars and sugar containing drugs

Natural Fruit Sugar

- Sugar from fruit is as natural as can be. That type of sugar is a lot better for you than refined sugars and even just straight granulated white sugar.
  - Glucose ("grape sugar") is also called *dextrose*
  - Fructose ("fruit sugar") is also called *laevulose*
  - Many fruits have their own sweet taste and they definitely don't need any added sugar!
  - Examples *Honey, Agave, Brown Rice Syrup, Date Sugar, Sorghum*

Sugars and sugar containing drugs

S.No	Name of the compound	Name of the drug	Botanical name
1	Arabin	Acacia	Acacia arabica
2	Guaran	Guar gum	Cyamopsis tetragonolobus
3	Glucose, fructose, sucrose	Honey	Apis mellifera
4	Tragacanthin, bassorin	Tragacanth	Astragalaus gummifer
5	Mannitol	Manna	Fraxinus ornus
6	Xylitol		

Sugars and sugar containing drugs

i. D-glucose (dextrose or α-D-glucopyranose)

a. Commercial preparation

- Even though glucose is abundant in many plant species
  - it is not extracted for commercial use.
  - It is prepared by enzymatic or mineral acid hydrolysis of starch.

Carbohydrates D-glucose cont.

b. Pharmaceutical uses include:

- ✓as sweetener
- ✓in the preparation of syrups and elixirs
- ✓for coating tablets and as tablet binder
- ✓in replacement therapy as a fluid and nutrient replenisher.
- ✓in the manufacture of
  - candy, carbonated beverages, ice cream and so on.

Carbohydrates & related products cont.

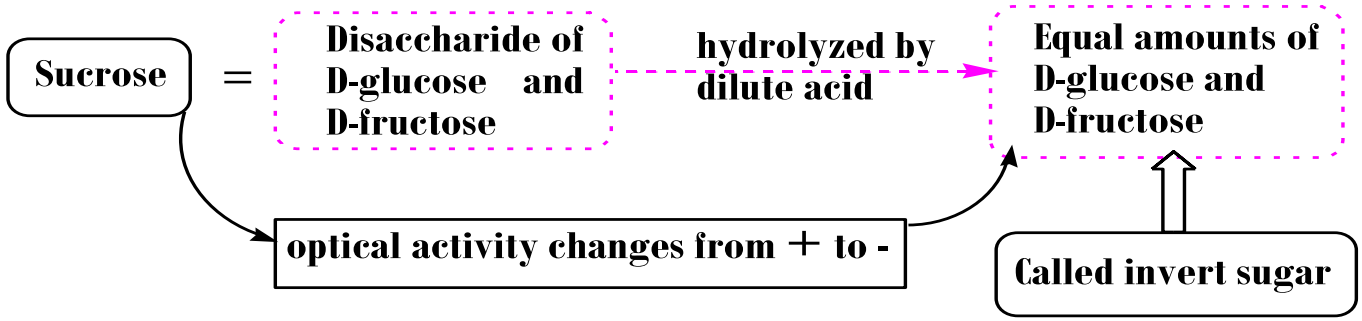
D-fructose (levulose)

- is present in all fruits and honey.
- Commercially obtained by
  - hydrolyzing inulin.
  - hydrolyzing sucrose.
  - enzymatic conversion of glucose.

- Uses:
  - As food for diabetic patients.
  - Infant feeding formula contains fructose.
  - As an ingredient for fructose injection; used as
    - ✓fluid, nutrient and electrolyte replenisher

Carbohydrates & related products cont.

Sucrose (saccharum) -non reducing sugar



- Important commercial sources include:
  - Sugar cane: *Saccharum officinarum* (fam. Poaceae)
  - Sugar beet: *Beta vulgaris* (fam. Chenopodiaceae)
- Use:
  - As standard sweetening agent for
    - ✓foods,
    - ✓syrups, and
    - ✓drug preparations.
  - As a demulcent and as a nutrient
  - Act as bacteriostatic agent and preservative at high conc.

Carbohydrates & related products cont.

Lactose is

- A disaccharide of D-galactose and D-glucose
- produced by crystallization
  - from the whey obtained in the manufacture of cheese.
- A reducing sugar.
- Commonly used as tablet diluents and infant food.

ii. Sugar related acids

a. Citric acid
b. Lactic acid
c. Tartaric acid

iii. Sugar related alcohols

a. Ethanol (alcohol)
b. Mannitol (D-mannitol)
c. Sorbitol (D-glucitol or D-sorbitol)

Carbohydrates & related products cont.

iv. Crude drug containing sugars

a. Honey

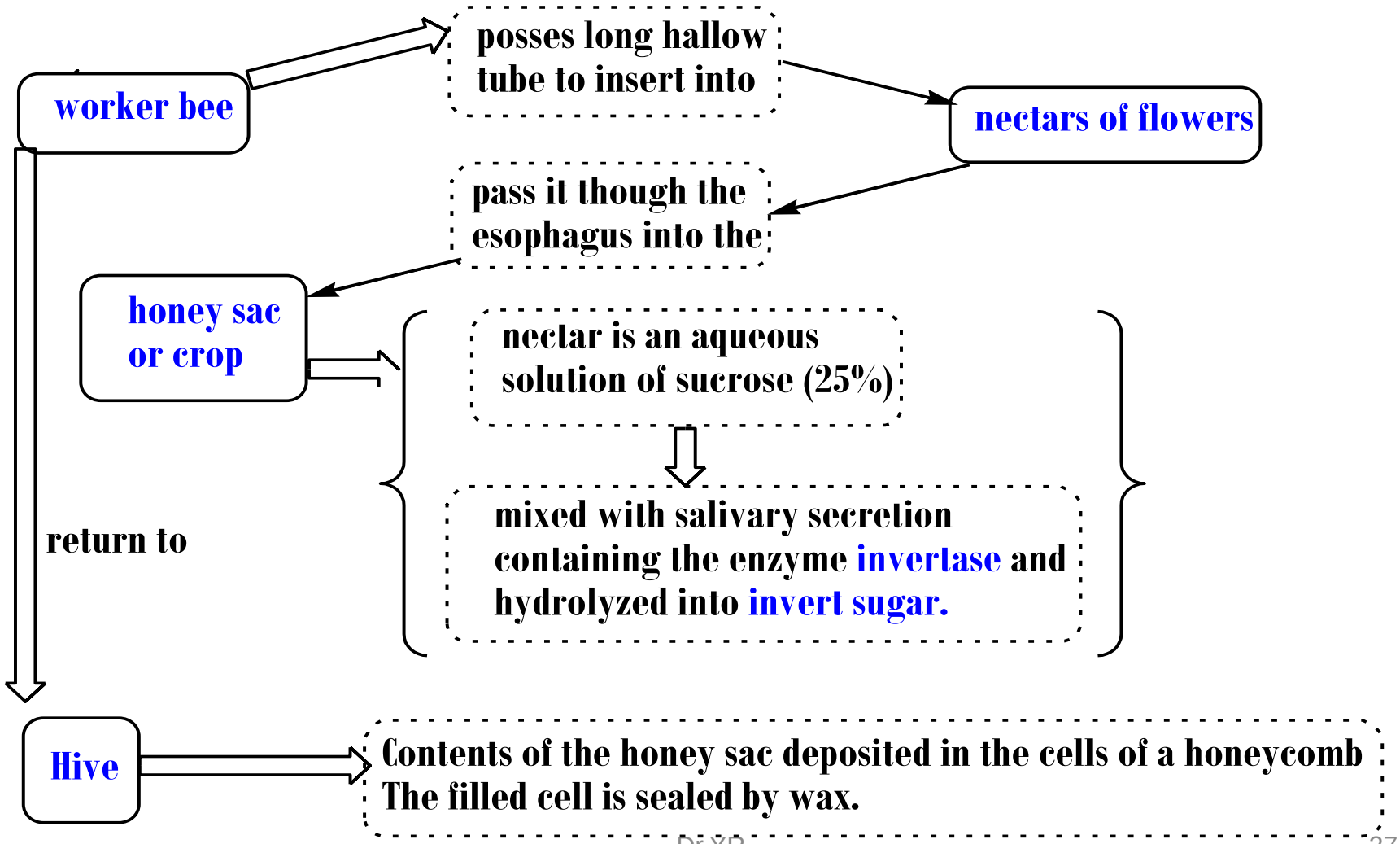
❖**Biological source**: Honey is a sugar secretion deposited in honey comb by the bees, *Apis mellifera, Apis dorsata* and other species of *Apis*, (family: Apidae)

❖It must be free from

- ➢foreign substances such as parts of insects, leaves and honeycomb but may contain pollen grains.

Carbohydrates - Honey cont.

Production and collection



Carbohydrates - Honey cont.

For collecting the honey,

- the bee hive is smoked to remove bees,
- the comb is cut and the honey collected either by
  - drainage or
  - expression.
    - ✓the honey could be contaminated by honeycomb.

Characters:

- Honey is
  - thick, syrupy, translucent liquid when fresh but
  - becomes opaque and granular through time due to crystallization of dextrose.
- Colour: It is pale yellow to reddish brown.
- Odour - Characteristic, pleasant
- Taste - Sweet and faintly acrid

Carbohydrates - Honey cont.

- Note however that:
  - The portion of sugar may vary depending upon the source of nectar and the enzymatic activity responsible for converting nectar into the honey.
  - Honey made from the nectar of some *Eucalyptus* spp. may have somewhat unpleasant odor and taste.
  - *Datura starmonium* may give poisonous honey.

Carbohydrates - Honey cont.

Chemical constituents:

- Honey contains
  - chiefly
    - ✓D-glucose (30-40%),
    - ✓fructose (40-50%) and
  - small amounts of
    - ✓sucrose (0.1-10%),
    - ✓pollen grains,
    - ✓pigments and
    - ✓volatile oil.
- Note: The pollen grains are used to trace back the floral source.

Carbohydrates - Honey cont.

- The other constituents of honey are
  - maltose
  - gum
  - traces of succinic acid
  - acetic acid
  - Dextrin
  - formic acid
  - colouring matters
  - enzymes(invertase, diastase, inulase) and
  - traces of vitamins.
- Proteins and pollen grains from various flowers are also found in honey.

Carbohydrates - Honey cont.

❖**Pharmacological activities and uses**

- Demulcent, (soar throat and tongue, and duodenal ulcer)
  - Similarly used for sore eyes
- Mild laxative (for constipation)
- Bactericidal and promotes healing
  - It is antiseptic and applied to burns and wounds.
  - Antibacterial properties duet to osmosis, high acidity etc.

Carbohydrates - Honey cont.

- Used for liver disorders (e.g. Hepatitis C)
- Sedative (for mild insomnia)
- Vermicide
- Emollient (important ingredient in cosmetics)

❖ **Common adulterants**

- Artificial invert sugar (glucose + fructose)
- Sucrose
- Commercial liquid glucose

- **Other Uses**:
  - As sweetening agent.
  - It is readily assimilated and hence is a good nutrient to infants and patients.
  - It is a common ingredient of several cough mixtures, cough drops and vehicle for ayurvedic formualtions.
  - Recently it is used in preparation of creams, lotions, soft drinks and candies.

Sugars and sugar containing drugs

- **Cane sugar** is sucrose which has been extracted from sugarcane, a tropical plant which produces **naturally** high concentrations of this sweet substance. Humans have been utilizing **cane sugar** in cooking for hundreds of years
- Raw sugar is cane sugar which has been minimally processed.

Compounds related to sugars

Caramel

- It is a concentrated aqueous solution prepared by heating sucrose and glucose until the sweet taste is destroyed and uniform brown viscous liquid results, by adding small quantities of sodium carbonate or mineral acid or small quantity of alkali, during heating.
- Description
  - Colour - thick or dark brown powder
  - Odour - Characteristic of burnt sugar
  - Taste - Pleasant bitter taste
  - Solubility - Soluble in water and in dil. alcohol

    Insoluble in organic solvents.
- Uses
  - It is used as colourant for food, confectionary, vinegars, liquors, malt beverages, tobacco flavouring and in elixirs, syrups and other oral liquid products in medicine.

Mannitol

- It is a sugar alcohol; that is, it is derived from a sugar by reduction, with a molecular weight of 182.17 g/mo, and a density of 1.52 g/mL.
- Mannitol and sorbitol are isomers, the only difference being the orientation of the hydroxyl group on carbon 2.
- Aqueous solutions of mannitol are mildly acidic and sometimes such solutions are treated to lower the pH.
- D-Mannitol has a solubility of 22g mannitol/ 100mL water (25°C), and a relative sweetness of 50 (sucrose=100).

- It melts between 165°-169°C (7.6 torr), and boils at 295°C.
- **Mannitol** is a white, crystalline organic compound with the formula ($C_6H_8(OH)_6$).
- This polyol is used as an osmotic diuretic agent and a weak renal vasodilator.
- It was originally isolated from the secretions of the flowering ash, called manna after their resemblance to the Biblical food, and is also referred to as **mannite** and **manna sugar**.
- In plants, it is used to induce osmotic stress.

Xylitol (from Greek)

- It is a sugar alcohol sweetener used as a naturally occurring sugar substitute.
- It is found in the fibers of many fruits and vegetables, including various berries, corn husks, oats, and mushrooms.
- It can be extracted from corn fiber, birch, raspberries, plums, and corn.
- Xylitol is roughly as sweet as sucrose with only two-thirds the food energy.
- Food manufacturers and processors add xylitol to foods like chewing gum, mints and some baked goods.

- Xylitol is categorized as a natural sugar substitute since it comes from natural sources, and the body produces sugar alcohols like xylitol naturally when it synthesizes sugars.
- Xylitol is sometimes referred to as birch sugar because manufacturers extract it from birch trees as well as berries, mushrooms and corn.
- Its only downside is that excess amounts of it will cause an upset stomach.

Properties of xylitol:

- 100 grams of xylitol contains 1405 kilojoules.
- One teaspoon (5 gm) of xylitol contains 9.6 kilocalories (Cal), as compared to one teaspoon of sugar, which has 15 Cal.
- Xylitol has virtually no aftertaste, and is advertised as "safe for diabetics and individuals with hyperglycemia."
- This tolerance is attributed to the lower impact of xylitol on a person's blood sugar, compared to that of regular sugars and also has a very low glycemic index of 13 (glucose has a GI of 100).

Complex Carbohydrates

Polysaccharides: Starch & Glycogen

Amylose

Amylopectin

Dietary fiber

Polysaccharides

Energy storage
Starches (amylose, glycogen, amylopectin) (glucans)
Levan and inulin (fructans)

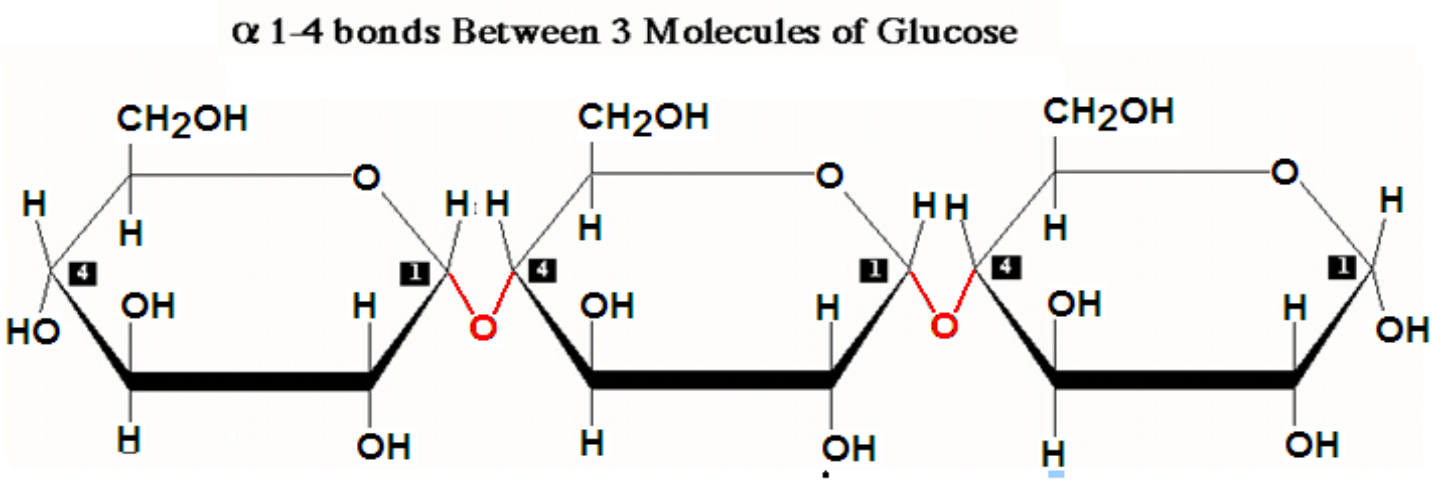
Structure - cellwalls
Cellulose (plants)
Chitin (yeast and arthropods)
Glucosaminoglucans (animals)
Murein (peptidoglucan, bacteria)

Polysaccharides

- 3 or more sugars linked together
- Complex sugars
- Important for energy storage

Examples:

Starch- (plants) found in leaves, tubers…
Glycogen- (animals) found in the liver and muscles
Cellulose- (plants) make up cell walls



Polysaccharides: Starch

- Biological Source:
  - Starch contains polysaccharide discrete granules obtained from the grains of
  - Maize(*Zea mays* Linn.)
  - Rice (*Oryza sativa* Linn.)
  - Wheat (*Triticum aestivum* Linn.) all belonging to

    Family : Graminae
  - Potato (*Solanum tuberosum* Linn.) belonging to

    Family : Solanaceae

Polysaccharides: Starch

- **Description**:
  - Colour - Rice starch is white coloured
    Wheat starch is cream coloured
    Potato starch is slightly yellowish
  - Odour - Odourless
  - Taste - Mucilagenous
- **Chemical constituents**:
  - Amylose - water soluble
  - Amylopectin - water insoluble
  - Amylose gives blue colour with iodine, while amylopectin yields Bluish black colouration.

Polysaccharides: Starch

- Starch granules are insoluble in water
- Starch occurs as granules with different shapes and size
  - (diameter = 2-100 µm) depending on the botanical source.
    - ✓Wheat starch: 2-23 µm
    - ✓Corn starch: 25-32 µm
    - ✓Potato starch 30-100 µm

Polysaccharides: Starch

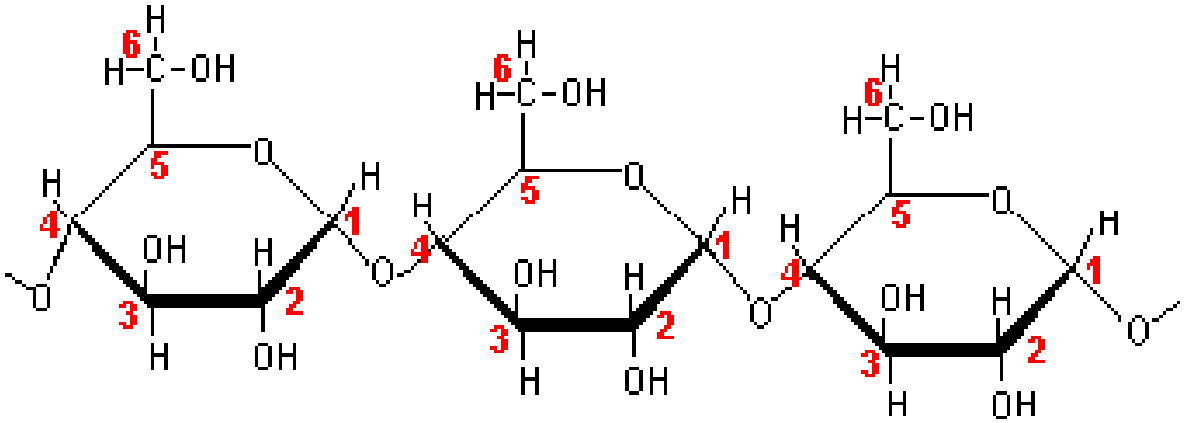
- 3,000 or more monosaccharides bound together
- **Plants** store glucose as **amylose** or **amylopectin**.
- Both are glucose polymers collectively called starch.
  - Amylose--straight chain polymer
  - Amylopectin--highly branched polymer

Polysaccharides: Starch

- Chemical composition:
  - Starch granules are composed of amylose and amylopectin
  - Variation in proportion of amylose and amylopectin is due to variation in properties of starch from different sources

	Amylose	Amylopectin
Average amount	20-25%	75-80%
Average no of glucose units	250-300	more than 1000
Structure	Linear non-branching	linear branching
Connectivity of glucose units	exclusively by α(1→4) linkage	α(1→4) linkage in the main chain and α(1→6) linkage at branching points
Water solubility	Water soluble	Water insoluble but swells in water
Color with iodine	deep-blue	blue-violet or purple

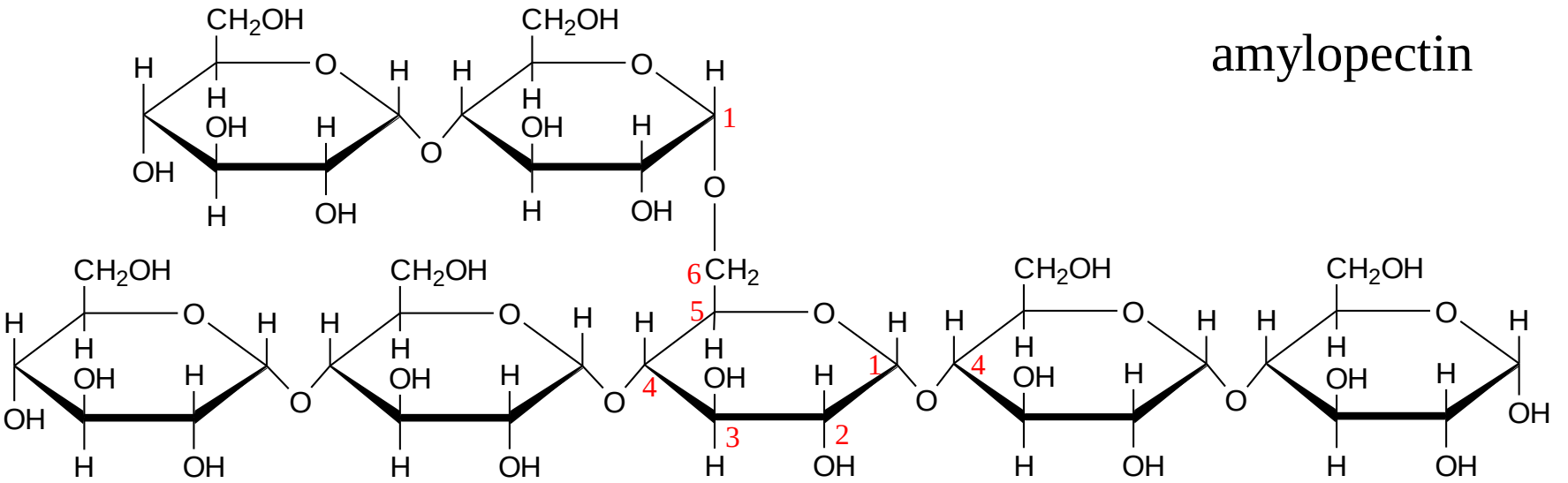
Polysaccharides - Starches



Amylose

- **Amylose** is a glucose polymer with α **(1→4)** linkages.
- The end of the polysaccharide with an anomeric C1 that is not involved in a glycosidic bond is called the **reducing end**.
- Glucose storage in **polymer** form **minimizes osmotic effects**.

Amylopectin



- **Amylopectin** is a glucose polymer with mainly $\alpha(1\rightarrow4)$ linkages, but it also has **branches** formed by α $(1\rightarrow6)$ linkages. Branches are generally longer than those shown in the diagram above.
- The branches produce a compact structure & provide multiple chain ends at which enzyme activity can occur.

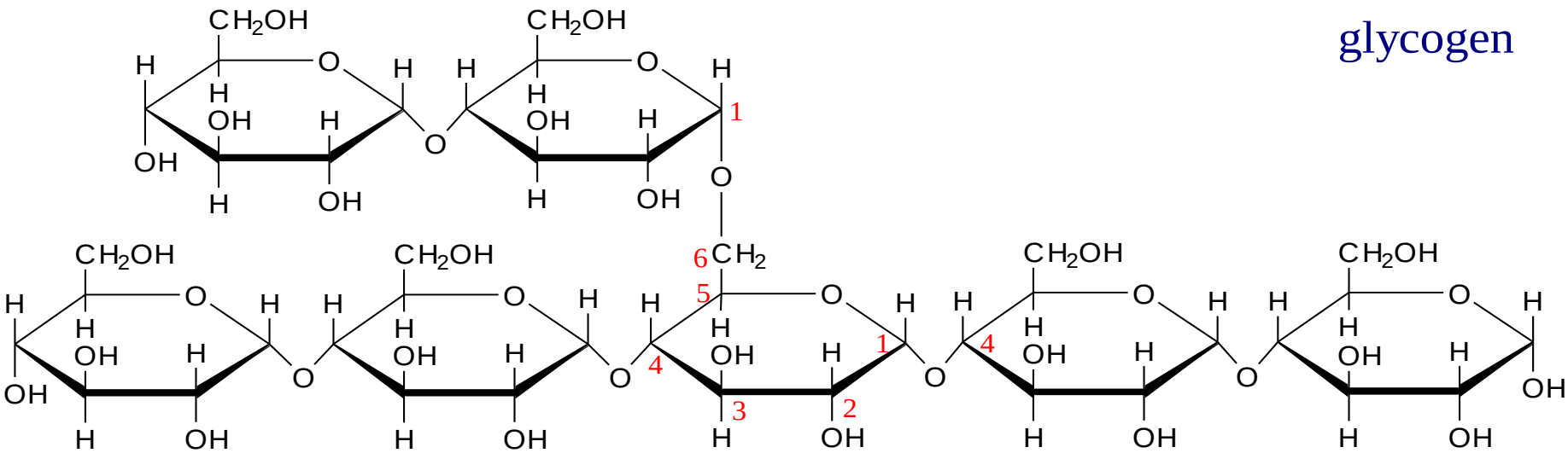
Polysaccharides - Starches

- **Starch is Used**
  - **Pharmaceutically as a**:
    - binder (because of amylose)
    - disintegranting agent in pills and tablets (because of amylopectin)
    - tablet filler
    - In the preparation of talcum powder
  - **Medicinally as:**
    - absorbent and emollient in dusting powders and in ointments
    - as antidote for iodine poisoning
    - As a nutritive, demulcent, protective and as an absorbent
    - As an antidote in iodine poisoning
    - As a starting material for the commercial manufacture of liquid glucose, dextrose and dextrin.
    - Industrially in sizing of paper and cloth.

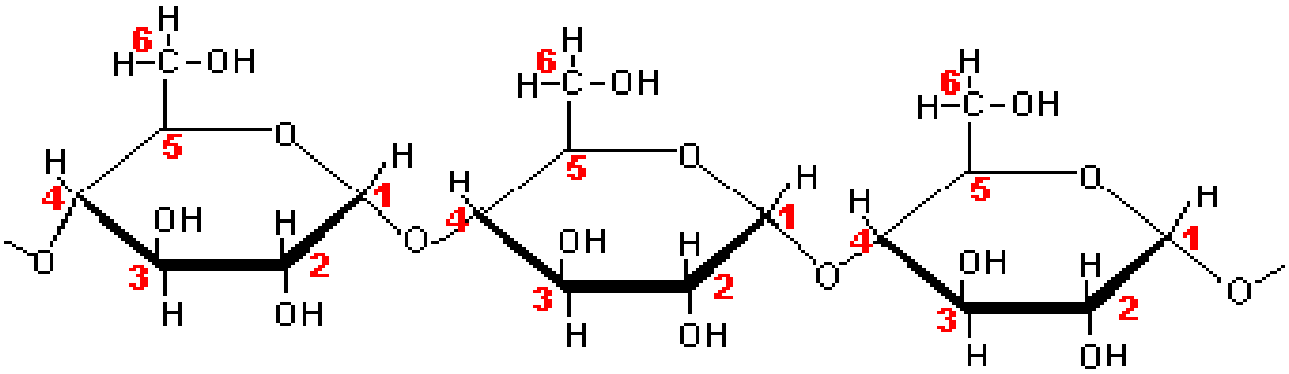
Glycogen

- **Glycogen**, the glucose storage polymer in **animals**, is similar in structure to amylopectin found in plants
- Glycogen has **more** α (1$\rightarrow$6) **branches** than amylopectin
- The ability to rapidly mobilize glucose is more essential to animals than to plants.
- The highly branched structure permits rapid glucose release from glycogen stores, e.g., in muscle during exercise.
- More sites for enzyme action
- Found in the liver and muscles

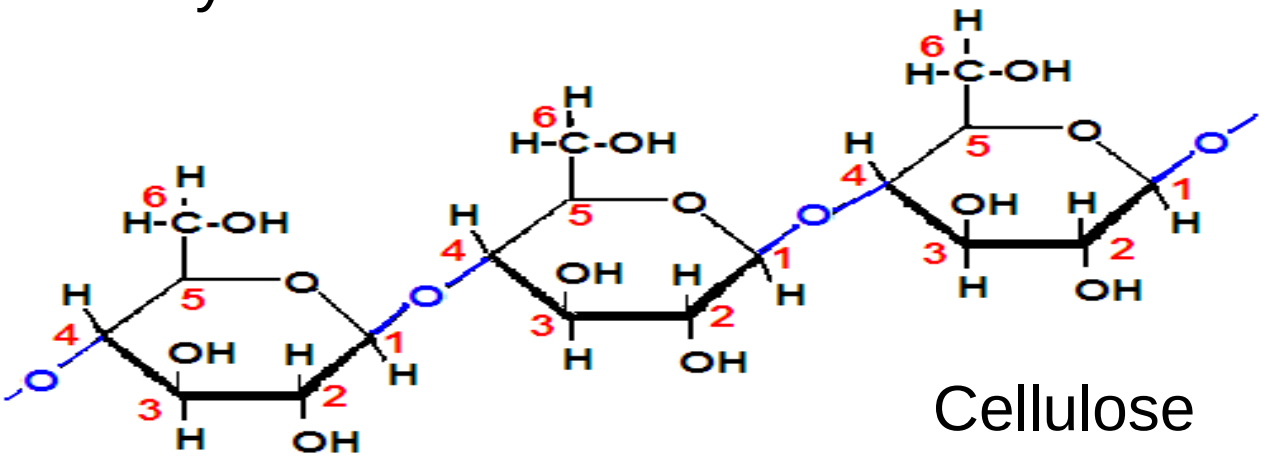
Glycogen



Starch and Cellulose



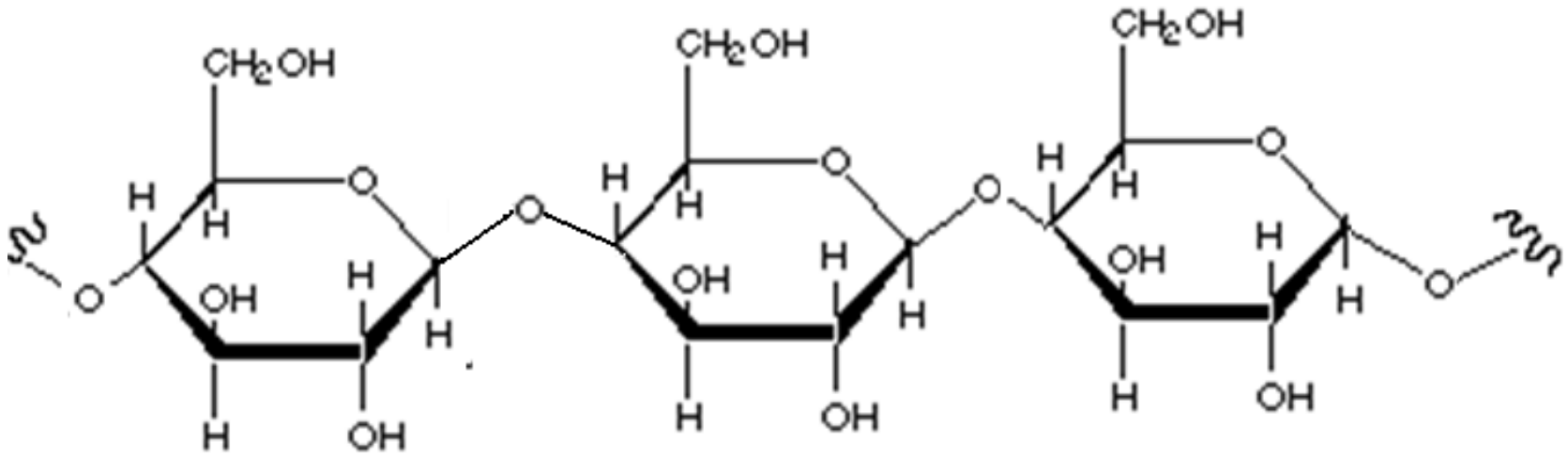
Amylose



Cellulose

The essential difference between amylose(starch) and cellulose is the glycosidic link between the saccharide units. Amylose has $\alpha(1\text{–}4)$ links. Cellulose has $\beta(1\text{–}4)$ links.

Cellulose



- Cellulose has β (1-4) links. Non branching.
- Cellulose is the major building component of plant cell walls
- Long chain of glucose molecules would be expected to be a great source of energy, but humans lack the necessary enzyme to digest cellulose
- The Endosymbiotic Protist in cow guts DOES have the enzyme
- the most abundant organic compound on earth
- the principal structural element of higher plant cell walls

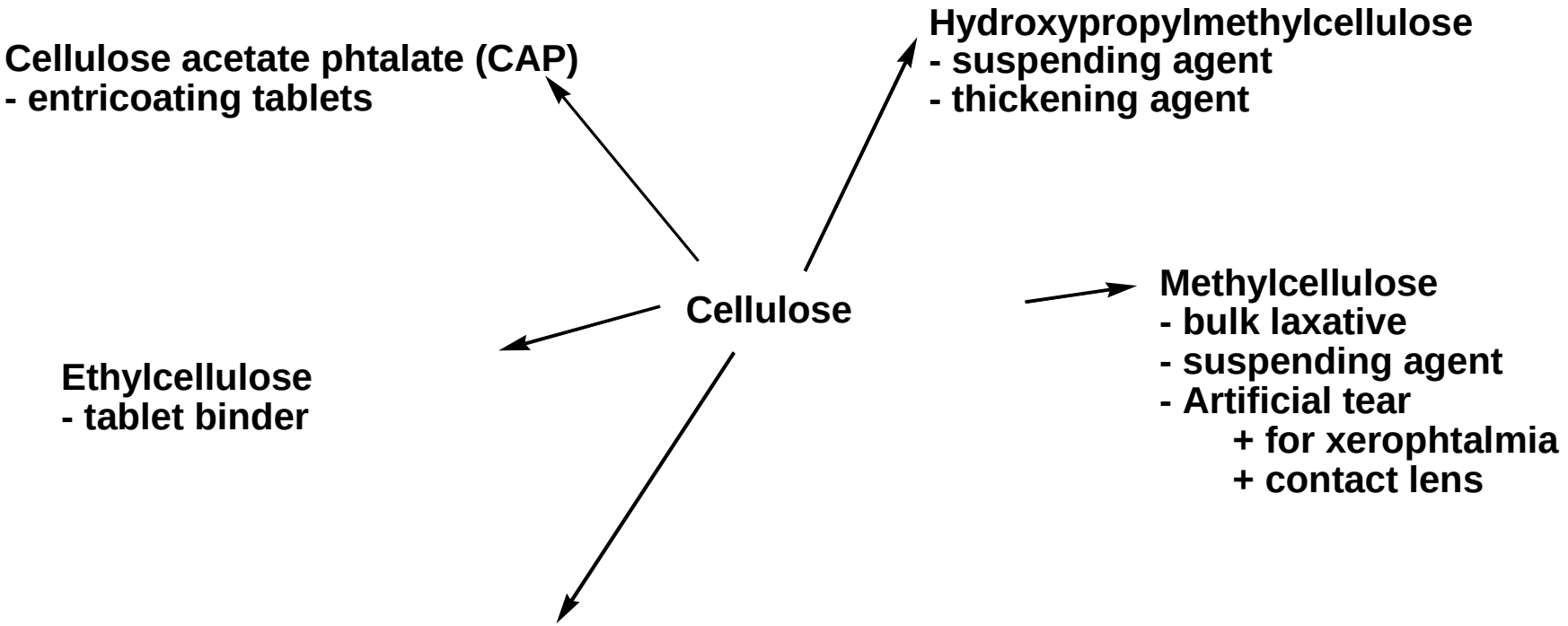
❖**commercially obtained from**

- ➢Cotton plant (*Gossypium* spp: Malvaceae)
  - ✓hair of the seed (98% cellulose)
  - ✓wood (40-50% cellulose)

❖**Purified cotton (absorbent cotton)**

- ➢obtained from cotton plant
- ➢It is used as
  - ▪surgical dressing
  - ▪for preparing absorbent gauze
  - ▪for textiles and
  - ▪in the manufacture of explosives

- Powdered cellulose
  - It is used as a self-binding tablet diluent and disintegrant
  - It is extensively modified



Inulin

❖**It** is a D-fructan (D-fructosan) in which

- 35-50 residues are linked in a linear manner by β (2→1) bonds and terminated by one glucose unit

- **Biological Source**:
  - It is a polysaccharide from the
  - Bulbs of Dehlia (*Inula helenium*)
  - Roots of Dandlion (*Taraxacum intybus*)
  - Burdock root (*Saussurea lappa*)
  - Chicory roots (*Cichorium intybus*)

  all belonging to family : Compositae

Description:

Colour - Colourless or white powder

Taste - Tasteless

Solubility- Soluble in hot water, Insoluble in organic solvents.

- **Chemical Nature**:

It is a polymer consisting of 35-50, 1-2 linked fructo-furanose units, terminated with one glucose molecule.

Inulin

- **Uses**:
  - Used as diagnostic agent.
  - In preparation of culture media as fermentative identifying agent for certain bacteria.
  - Manufacture of fructose, by using as enzyme inulase.
  - Ingredient of diabetic bread.
  - not hydrolysed in the GIT.
  - It is used
    - for evaluation of renal function
    - also as dietary fiber
  - not metabolized if administered iv
  - It undergoes glomerular filtration and
    - ✓it is neither absorbed nor excreted in the tubule

Xanthan gum

- **Biological Source**:
  - It is a polysaccharide produced by certain species of bacteria.
  - It is produced by pure fermentation of glucose using the bacterium named as *Xanthomonas compestris*, which is non-toxic and non pathogenic for human beings.
- **Description**:
  - Colour - Cream coloured powder
  - Odour - odourless
  - Solubility- Soluble in cold and hot water and gives highly viscous solution.

Xanthan gum

- **Chemical nature**:
  - Xanthan gum consists of chain composed of D-glucose, D-mannose and D-glucuronic acid with short side chains.
- **Uses**:
  - It is a very good emulsifying agent, stabilizer and thickener.
  - It is used in pharmaceutical and cosmetic industries.
  - It is extensively used in food industry for dairy products, salad dressings and canned products.

Dextrin

- **Biological source**:
  - It is produced by incomplete hydrolysis of starch with dilute acids or by heating dry starch.
- **Description**:
  - Colour - Pale yellow coloured
  - Odour - Characteristic
  - Solubility - Soluble in boiling water
    Slightly soluble in cold water
    Insoluble in alcohol and ether
- Uses :
  - Used as tablet excipient
  - In preparation of match sticks and explosives.

Dextran

- It is polysaccharide produced by growing bacteria on the substrate sucrose, containing a backbone of D-glucose units linked predominantly α–D(1-6).
- Dextrans are water soluble and insoluble in alcohol.
- Dextrans molecular weight varies from 40,000 to 75,000.
- It is used as a plasma volume expander.
- Dextran 40 is also used as blood flow adjuvant.

Gums and Mucilages

- **Gums and mucilages**
  - have similar constitutions and
  - on hydrolysis yield a mixture of sugars and uronic acids.
  - are prrcipitated from solutions by alcohol and lead subacetate.
  - Pharmaceutically important gums:
    - Acacia gum, Gum tragacanth
  - Pharmaceutically important mucilages
    - Agar, Algin

Gums and Mucilages

- **Gums**
  - considered to be **pathological products** formed upon injury of the plant or owing to unfavorable conditions such as drought
  - usually serve as a protective
- **Mucilages**
  - **normal products of metabolism**
  - usually serve as storage materials, water-storage reservoirs, or may provide protection for germinating seeds.

Gums and Mucilages

- Gums are either hydrophobic or hydrophyllic high molecular weight molecules, usually with colloidal properties.
- In appropriate solvent or swelling agent, they produce gels, highly viscous suspensions or solutions.
- Depending upon types of occurance, gums can be classified as

Gums

Natural gums

1. Seaweed gum

Eg.,agar, algin, carrageenan, lamanarian

2. plant exudates

E.g.,acacia, tragacanth, karaya, ghatti

3. Seed gums

E.g.,guar gum, isabgol, tamarind, locust bean gum

4. Plant extracts

E.g.,pectin , larch, arabinogalactan

5. Animal extract E.g.,chitin.

Prepared gums

1. Biosynthetic gums xanthan, scleroglucan, dextrans
2. Starch and its derivatives dextrins
3. Cellulose derivatives.

Gums and Mucilages

- Isapgol
- Agar
- Acacia
- Tragacanth
- Carragenan

Gums and Mucilages

Isapgol (Ispaghula, Isabgul, Flea seed)

- **Biological source**:
  - It consists of dried seeds of the plant known as *Plantago ovata*(family: Plantaginaceae)
- **Description(macroscopic characters)**:
  - Colour - Pinkish-grey or brown
  - Odour - None,
  - Taste - Bland, mucilaginous
  - Shape - It is ovate

Isapgol

- **Chemical constituents**:
  - It contains mucilage which is present in the epidermis of the seeds.
  - Chemically, it consists of pentosan and aldobionic acid.
  - The products of hydrolysis are xylose, arabinose, galacturonic acid and rhamnose.
- **Uses:**
  - Seeds are used as demulcent, laxative, emollient and in the treatment of chronic constipation, amoebic and bacillary dysentery.
  - Mucilage of isapgol is used in the preparation of tablets and also as a stabilizer in the ice-cream industry.
  - The mucilage has property of glairiness and stringiness, which is desired in certain cosmetic formulations.

Gums and Mucilages

Agar

- **Biological source**:
  - It is dried gelatinous substance obtained from *Gelidium amansii* (family: Gelidaceae) and other species of red algae like, *Gracilaria* (gracilariaceae) and *Pterocladia*(Gelidaceae)
- **Description**:
  - Colour - it is yellowish-grey or white to nearly colourless
  - Odour - Odourless
  - Taste - Mucilagenous
  - Shape - It is found in various forms like stripes, sheets, flakes or coarse powder.

Agar

- **Chemical constituents**:
  - Agar consists of two different polysaccharides names as agarose and agropectin.
  - Agarose is responsible for gel strength of agar and is composed of D-galactose and 3,6 anhydrous L-galactose units.
  - It contains about 3.5% cellulose and 6% of nitrogen containing substance.
  - Agaropectin is responsible for the viscosity of agar solutions.
  - It is believed to be a sulphonated polysaccharide in which galactose and uronic acid units are partly esterified with sulphuric acid.
- **Uses**:
  - Agar is used as an emulsifying agent and bulk laxative.
  - It is used in the preparation of jellies, confectionery items and in preparation of biological culture medium(1% solution).
  - Bulk laxative in chronic constipation
  - For gel chromatography

Gums and Mucilages

Acacia (Gum acacia, Gum arabic)

- **Biological source**:
  - It is the dried gummy exudation from the stem and branches of *Acacia arabica*(family: Leguminosae)
- **Description**:
  - Colour - Tears are cream-brown to red in colour, while powder is light brown in colour
  - Odour - Odourless
  - Taste - Bland and mucilagenous
  - Shape - Irregular tears of varying sizes.
  - Solubility - Soluble in water, in soluble in alcohol.

Acacia

- **Chemical constituents**:
  - It consists principally of arabin, which is complex mixture of calcium. Magnesium and potassium salts of arabic acid.
  - Arabic acid on hydrolysis gives
    - L-arabinose
    - L-rhamnose
    - D-galatose and
    - D-glucuronic acid.
  - It also contains an enzyme oxidase, and peroxidase.
- **Uses**:
  - It is used as demulcent. (added in formulations for cough, diarrhea and throat problems)
  - emollient (to cover burns, sore nipples etc)
  - It is also administered intravenously in haemolysis.
  - In the form of mucilage it is used as suspending agent.
  - It is a good emulsifying agent(for oil in water emulsions).
  - It is a good binding agent and is used in the preparation of lozenges, pastilles and compressed tablets.
  - used as adhesive commercially.

Gums and Mucilages

Tragacanth

- **Biological source**:
  - It is the dried gummy exudation obtained by incision from stems and branches of *Astragalus gummifer* and other species of Astragalus, (family:Leguminosae, fabaceae)
- **Description**:

Colour	-	The flakes are white or pale yellowish-white.
Odour	-	odourless
Taste	-	mucilagenous

Tragacanth

- **Chemical constituents**:
  - It contains two fractions of which one is soluble in water.
  - The water soluble portion is known as tragacanthin.(8-10%) of the gum.
  - Water insoluble portion is known as bassorin (60-70%).
  - Tragacanth contains about 15% of methoxy group which swells in water.
  - The products of hydrolysis are galacturonic acid, D-galactopyranose, L-arabino-rhamnose and D-xylopyranose.
  - is resistant to acid hydrolysis & is preferred in acidic preparations.
- **Uses:**
  - It is used as demulcent, and as emollient in cosmetics.
  - It is used as thickening, suspending and as an emulsifying agent.
  - It is used along with acacia as suspending agent.
  - Mucilage of tragacanth is used as a binding agent in the tablets and also as excipient in pills. Tragacanth powder is used as an adhesive.
  - In spermicidal jellies, as a stabilizer in ice-creams(concentration 0.2-0.3%).

Gums and Mucilages

Carragenan

- **Biological source**:
  - It is sulphated polysaccharide extract of the sea weed called carrageen; or irish moss, the red-algae obtained from *Chondrus cripus*
- **Family**: Rhodophyceae
- **Chemical constituents**:
  - Depending upon the position of sulphate and presence or absence of anhydrogalactose, two types of carragenan are known kappa(k) and Lambda(λ).
  - k-carragenan contains D-galactose, 3,6 anhydrous-D-galactose and ester sulphate group. It is a good gelling agent.
  - λ-carragenan contains D-galactose and its mono and disulphate esters. It is non-gelling agent and is a good thickener.

Carragenan

- **Description**:
  - It is water soluble colloid, dried to form free flowing powder.
  - Depending upon the method of preparation, two forms of drugs are available.
  - One which is extracted with cold water gives viscous solution in water, and the other extracted with hot water forms gel on cooling.
- **Uses**:
  - It is used as emulsifying agent, stabilizing agent, gelling and viscosity builder in food products.
  - Tooth pastes, creams, lotions and other cosmetic products are prepared by using carragenan
  - In food industry it is utilised in milk products, ice creams, chocolate, puddings, jams, gels in the concentration of 0.5-1.0%.

Gums and Mucilages

- **Algin (Sodium alginate)**
  - is extracted from different strains of *brown seaweed* by the use of dilute alkali
  - is chiefly obtained from *Macrocystis pyrifera* (Lessoniaceae)
  - Used as
    - thickening, suspending, and emulsifying agent
    - tablet binder
    - food (ice cream)
    - Commercially, in cosmetics and textile

Carbohydrate Functions: Storage

- The main storage polysaccharides are starches and glycogen. While plants use starch as their storage polysaccharides, animals use glycogen.
- When the body has a high glucose concentration, the pancreas releases insulin, which converts glucose into glycogen and stores it in the liver.
- When the glucose concentration is low, the hormone glucagon converts glycogen back into glucose.
- Glycogen is the primary energy reserve in human beings . Metabolism of glucose provides the energy necessary for our bodies to function and carry out daily activities.
- When it is broken down into glucose and oxidized, ultimately to CO_2 and H_2O, through cellular respiration, large amounts of energy are released.

Carbohydrate Functions: **Energy Sources**

- During metabolism animals break down carbohydrates to carbon dioxide and water vapor.
- Monosaccharides and dissaccharides break down quickly and provide quick energy sources.
- Starches take longer to metabolize but the end products are the same.
- Human beings cannot break down cellulose, since we lack the appropriate enzyme to breakdown the β 1-4 linkage

The End